Streamlined Bakery Management: A Dedicated System for Cake Customization and Sales Monitoring

A Capstone Project Presented to the Faculty of the College of Information Technology Education

PHINMA University of Iloilo

In Partial Fulfillment of the Requirements for the degree Bachelor of Science in Information Technology

By:

Senosa, Kristle Reign C. - BSIT 3-7

Solamillo, Justin Brian A. - BSIT 3-7

Roa, Althea Marie P. - BSIT 3-7

Tibajares, Noli Jr. A. - BSIT 3-7

Palcullo, Julliene Kathryn T. - BSIT 3-7

Data Informatics Track

Second Semester 2025-26

CHAPTER I

INTRODUCTION

1.1 Background of the Study

In the modern business landscape of the Philippines, micro, small, and medium enterprises (MSMEs) were increasingly pressured to adapt to digital transformation. For local bakeshops, the traditional "brick-and-mortar" approach and absolute reliance on social media channels proved insufficient to meet the expectations of a tech-savvy customer base. Modern consumers prioritized convenience, actively seeking platforms where they could seamlessly browse products, customize intricate designs, track order fulfillment, and settle payments securely without repetitive manual inquiries.

Mrs. Brave’s Cakes operated as a homegrown pastry business owned and managed by Mrs. Shamelyn Tapang since 2020. Specializing in customized cakes for weddings, birthdays, and significant milestones, the business built a loyal following through its creative designs and quality products. However, despite its market growth, the administrative, delivery coordination, and storefront operations remained heavily manual. Mrs. Tapang utilized a Facebook page as her primary digital storefront. While social media served as an excellent tool for marketing, it lacked the structured data environment necessary for managing the complex variables of custom cake production, localized delivery routing, and simultaneous walk-in transactions.

The process of creating a custom cake involved tracking specific attributes such as tier sizes, frosting types, edible prints, toppers, and rush timelines. When these details were gathered through fragmented chat threads on Facebook Messenger, the risk of miscommunication remained high. A single overlooked detail in a long text conversation often led to production errors, wasted ingredients, and customer dissatisfaction. Furthermore, as a sole proprietor, Mrs. Tapang managed every aspect of the business—from physical baking to financial tracking and dispatching independent riders. The lack of a centralized ecosystem meant that raw ingredient inventory, physical walk-in sales, and delivery handoffs were handled through disjointed manual observation, which was highly time-consuming and prone to human error.

To address these critical operational gaps, this study will develop the Streamlined Bakery Management: A Dedicated System for Cake Customization and Sales Monitoring. By migrating Mrs. Brave's Cakes from a chat-based workflow to a dedicated, unified web ecosystem, the platform will introduce an interactive customer cake customizer with real-time price calculations, an automated inventory-linked administrative dashboard, an on-site Point of Sale (POS) system for walk-in clients, and a dedicated coordinates portal for external delivery riders. Backed by real-time synchronization via Google Firestore, this system will minimize administrative bottlenecks, will prevent scheduling overbooking through a dynamic calendar control featuring admin-defined blackout dates, and will completely streamline the entire ordering-to-delivery pipeline.

1.2 The Existing Manual System

To fully evaluate the necessity of the proposed system, the baseline workflow of Mrs. Brave's Cakes was thoroughly analyzed. The operational structure relied heavily on three unintegrated components: a Facebook Business Page, a personal GCash account, and a physical paper ledger.

The existing manual system followed a fragmented process flow across four distinct phases:

• The Inbound Phase: A customer initiated an inquiry via Facebook Messenger. Mrs. Tapang had to manually respond to every message, constantly repeating base pricing matrices, flavor choices, and availability rules to multiple clients simultaneously.

• The Specification Phase: For custom orders, a lengthy back-and-forth exchange of reference images and design notes occurred. Mrs. Tapang had to mentally calculate the total costs based on fluctuating ingredient prices and design complexity, which frequently led to inconsistent, inaccurate, or undercharged pricing.

• The Fulfillment and Payment Phase: Once an agreement was reached, payments were routed to a personal GCash account. Mrs. Tapang had to manually cross-reference SMS notifications on her personal phone to verify payments before writing the entry into a physical logbook. Walk-in customers were handled separately via physical cash, with transactions recorded randomly or omitted entirely, which severely distorted the bakery's true daily sales metrics and analytics.

• The Dispatch Phase: On the day of delivery, Mrs. Tapang had to coordinate with external riders by manually copying and pasting addresses or explaining geographic landmarks via chat. Because there was no direct data loop between the business, the customer, and the driver, customers were left completely unaware of their delivery status, while the owner had to constantly send manual, real-time updates.

This manual architecture presented critical failure points: physical logbooks were constantly vulnerable to loss or damage, automated scheduling conflicts occurred during peak seasons due to the lack of a hard booking limit, and manual delivery dispatch invited severe operational delays. These bottlenecks collectively restricted the business's scalability and inflated daily operational stress.

1.3 Statement of the Problem

Mrs. Brave's Cakes faced several interconnected operational and administrative challenges that hindered its growth, compromised its daily efficiency, and increased its vulnerability to human error. Despite the market success of the homegrown bakery, absolute reliance on manual procedures across isolated channels caused frequent business bottlenecks. Specifically, this study aimed to address the following critical problems:

- **Inefficiency in Customization Tracking:** Relying on unstructured Facebook Messenger chat threads for complex, multi-tiered cake orders made it exceptionally difficult to systematically retrieve and finalize client design requirements. This lack of a structured input form frequently led to communication gaps regarding tier sizes, icing types, and fondant accents, which resulted in material waste and costly production modifications.
- **Manual and Inconsistent Pricing Calculation:** There was no standardized tool to dynamically compute the total cost of custom cake configurations, add-ons, or rush order premiums. Pricing was calculated manually by the owner based on mental estimates or loose recall, leading to delayed response times and inconsistent, non-standardized quotes that risked undercharging the business or confusing the consumer.
- **Unregulated Scheduling and Kitchen Overbooking Risks:** Without a centralized, real-time booking ledger, tracking pickup dates, delivery deadlines, and production

schedules relied heavily on hand-written notes. This increased the risk of overbooking the kitchen's manual production capacity during peak milestone seasons or completely missing time-sensitive event deadlines due to the lack of hard calendar boundaries.

- **Fragmented Financial Tracking and Omitted Walk-in Metrics:** The reliance on a physical paper ledger to track revenue and raw ingredient expenses made generating weekly, monthly, or seasonal financial statements incredibly tedious. Furthermore, physical walk-in cash transactions were frequently unrecorded or tracked completely separate from online orders, creating a disconnected data stream that distorted the owner's visibility over true net profit margins and overall business financial health.
- **Reactive Inventory Management:** Raw material stock levels were monitored exclusively through manual pantry inspections and erratic notes. The absence of an active inventory tracking mechanism prevented the owner from receiving timely low-stock indicators, resulting in the late discovery of missing baking supplies mid-production or the unnecessary over-purchasing of redundant stock.
- **Disjointed Delivery Logistics and Communication Blindness:** Once an order left the bakery, delivery routing and coordination with third-party riders were handled via loose text messaging or external map lookups. Because there was no unified data link passing customer location data directly to the courier, the owner faced severe delays in dispatching drivers, riders struggled to locate drop-off points, and customers suffered from total status blindness regarding the real-time fulfillment state of their orders.

1.4 Objectives of the Study

General Objective: The primary goal of this research is to design, develop, and implement a

"**Streamlined Bakery Management: A Dedicated System for Cake Customization and Sales Monitoring**" for Mrs. Braves Cakes. This unified digital ecosystem aims to replace the fragmented manual and social-media-dependent workflow by centralizing interactive online orders, physical walk-in sales, real-time capacity scheduling, and delivery dispatch into a single, cohesive platform.

Specific Objectives: To achieve the general objective, the study specifically aims:

1. **To develop an Interactive Customization Module** where customers can configure specific cake attributes (such as tiers, size, icing type, and toppers), upload visual reference photos, and view dynamic, real-time pricing updates before checkout.
2. **To implement a Dual-State Production Calendar** within the admin dashboard that allows the owner to manually approve custom orders, set blackout dates, and automatically halt online pre-made checkouts on fully booked days to prevent kitchen over-capacity.
3. **To construct an on-site Point of Sale (POS) Module** that allows the administration to process and record physical walk-in transactions alongside online orders, complete with thermal printer compatibility for physical receipts, ensuring that daily financial metrics accurately reflect total business revenue.
4. **To deploy a Dedicated Rider Dispatch Interface** that securely transmits static customer location coordinates to external couriers via generated links, allowing riders to utilize native mapping applications for routing and instantly update order fulfillment statuses.
5. **To integrate a Centralized Inventory and Analytics Dashboard** where the administrator can manually track ingredient stocks, monitor available pre-made cake quantities, and generate automated visual reports detailing daily sales, expenses, and net

profit margins.

6. **To establish a Secure Payment and Receipting System** utilizing the PayMongo API (for online GCash/Maya transactions) alongside cash-handling options, allowing customers to download digital e-receipts directly from the platform while simultaneously auto-dispatching copies to their registered email (e.g., Gmail) to ensure secure and accessible transaction records.

1.5 Significance of the Study

The completion and implementation of this study will provide significant, measurable benefits to the following specific stakeholders:

- **The Administrator (Mrs. Shamelyn Tapang):** This study will be most significant for the business owner. The system will act as a comprehensive digital assistant, eliminating the heavy lifting of manual bookkeeping, fragmented chat inquiries, and unrecorded walk-in sales. By centralizing online orders, the on-site POS, and delivery dispatch, it will minimize administrative stress, will prevent calendar overbooking through hard capacity limits, and will eliminate hours of manual computation, allowing the owner to focus entirely on culinary production and business scaling.
- **The Customers:** For the patrons of Mrs. Brave's Cakes, this system will offer a highly professional, transparent, and convenient e-commerce experience. They will gain the capability to visually configure their orders, receive instant pricing computations, safely process online payments via PayMongo, download digital e-receipts, and track their order status without waiting for manual, delayed replies on social media.
- **The Delivery Couriers (Riders):** By utilizing the dedicated Rider Interface, external

drivers will benefit from clear, digital dispatch instructions. The system will minimize communication errors by providing exact, static map coordinates and digital order summaries, making destination routing faster, safer, and more accurate.

- **The Researchers:** This project will serve as a comprehensive application of the researchers' academic skills in Full-Stack web development, NoSQL Database Management, and UI/UX design. It will provide invaluable real-world experience in resolving complex operational, financial, and logistical bottlenecks for a local Philippine MSME.
- **Future Developers and Researchers:** This manuscript and system architecture will serve as a detailed structural reference for future IT students developing integrated e-commerce, POS, and dispatch management systems for specialized food-based or inventory-dependent enterprises.

1.6 Scope and Delimitation

Scope

The proposed project will be a unified, web-based management ecosystem specifically designed for Mrs. Brave’s Cakes. The system will be structurally divided into three primary operational modules.

- **The Customer Interface:** This module will feature a product catalog, an interactive custom cake builder with dynamic price calculations, PayMongo payment gateway integration (for GCash and Maya transactions), and real-time order status tracking.
- **The Administrator Dashboard:** This module will equip the owner with a physical walk-in Point of Sale (POS) system, a manual inventory tracking ledger for raw

ingredients and pre-made stock counts, visual financial analytics charts, and dual-state production calendar to manage custom order approvals and define business blackout dates.

- **The Rider Dispatch Portal:** This module will function as a dedicated web interface for external couriers to view assigned delivery assignments, access static customer coordinates, and synchronize fulfillment statuses.

The application will be engineered using React.js for the responsive web frontend, a Python backend framework for centralized data handling, and Google Firestore (NoSQL) for real-time cloud database synchronization.

Delimitations

To maintain the technical feasibility and strict focus of the study, the system will be bounded by the following specific constraints:

- **No Live GPS Tracking:** The Rider Portal will rely purely on static geographic coordinates pinned by the customer during checkout. It will generate a parameterized web link to launch native mapping applications (such as Google Maps) but will not feature real-time, continuously moving GPS tracking of the rider's physical device.
- **No Automated Hardware Integrations:** The inventory system will function strictly as a digital manual ledger. It will not integrate with IoT physical hardware, barcode scanners, or automated weight sensors to log raw ingredients; all raw material stock adjustments will require manual administrative input.
- **No Native Mobile Application:** The system will be developed exclusively as a responsive web-based application accessible through desktop and mobile browsers. It

will not be compiled, packaged, or deployed as a native mobile application (such as an Android APK or iOS IPA) on any public app store.

- **No Third-Party Fleet API Integration:** While the system will facilitate the dispatching of external drivers, it will not connect directly to the backend automated APIs of corporate on-demand logistics fleets (such as GrabExpress, Lalamove, or Borzo). Rider assignment and link delivery will be handled manually by the administrator via external messaging platforms.

1.7 Definition of Terms

To ensure clarity and uniform understanding, the following terms are defined operationally as they are used in this study:

- **Add-ons:** Operational: Extra customizable cake decorations (e.g., fondant characters, custom toppers, or edible prints) selected by the customer that dynamically increase the computed base price of the product.
- **Administrator:** Operational: Refers to the business owner, Mrs. Shamelyn Tapang, who holds centralized access to the backend system, including the POS, inventory management, calendar scheduling, and rider dispatch controls.
- **API (Application Programming Interface):** Operational: In the context of this study, it specifically refers to the PayMongo API, which is integrated into the system to securely process online GCash and Maya payments.
- **Blackout Dates:** Operational: Specific calendar dates manually locked or disabled by the administrator to prevent automated online checkouts when the kitchen's production capacity for custom cakes is fully booked.

- **E-Receipt:** Operational: The digital proof of purchase automatically generated by the system upon a successful transaction, which can be downloaded directly from the platform or received via the customer's registered email (e.g., Gmail).
- **Inventory Monitoring:** Operational: The digital ledger system used by the administrator to **manually** track, deduct, and update the stock levels of raw baking supplies and available pre-made cakes, replacing physical logbooks.
- **NoSQL (Google Firestore):** Operational: The non-relational, cloud-based database structure utilized by the system to store, retrieve, and synchronize flexible data—such as custom cake orders, coordinates, and inventory counts—in real time.
- **Point of Sale (POS):** Operational: The dedicated software module within the admin dashboard used to process, calculate, and record physical walk-in cash transactions, designed with compatibility for on-site thermal receipt printers.
- **Rider Dispatch Interface (Rider Portal):** Operational: The dedicated web environment accessed via a generated link, used by external delivery couriers to view static customer map coordinates, contact details, and update the final delivery status.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter provides a comprehensive review of various literatures, studies, and existing systems—both local and foreign—that are relevant to the development of the "**Streamlined Bakery Management: A Dedicated System for Cake Customization and Sales Monitoring**" for Mrs. Braves Cakes. The researchers gathered information from academic journals, books, and existing software to provide a theoretical and technical foundation for the study.

2.1 Foreign Literature

2.1.1 Digital Transformation in Micro-Enterprises. According to Miller (2022) in the book *Modern E-commerce for Food SMEs* (ISBN: 978-3-16-148410-0), the digitalization of small-scale food businesses is no longer a luxury but a fundamental requirement for survival in a post-pandemic economy. Miller argues that manual order tracking in bakeries leads to a significantly higher rate of administrative errors compared to automated systems. The literature emphasizes that a dedicated customization interface allows customers to feel more "in control" of their purchase, which leads to higher customer satisfaction and repeat business. For an artisanal business, the shift from social media chats to a structured web platform represents a transition from informal to formal business operations. This formalization is necessary to build a brand that can survive long-term market fluctuations.

2.1.2 The Efficiency of Admin-Led Inventory Models. According to Dr. Robert Hales (2023) in the *Journal of Small Business Technology* (ISSN: 2192-3078), fully automated inventory systems often fail in artisanal businesses because software cannot account for "kitchen variables" like accidental spills, minor ingredient variations, or the "by-the-eye" measuring common in custom baking. Hales argues that an **Admin-Led Inventory Model**, where the system provides a centralized digital dashboard but the owner manually verifies and updates stock, is much more effective for sole proprietors. This "Hybrid" approach ensures data accuracy without being overly rigid. It allows the owner to remain the primary decision-maker while using the software as a highly organized digital ledger.

2.1.3 User Interface (UI) Design for Product Customization. In the book *Digital Consumer Behavior* (ISBN: 978-1-108-47084-1), Dr. Elena Rossi (2022) discusses the "Paradox of Choice." She explains that while customers want to customize their products, a messy interface leads to "order abandonment." Rossi suggests that a step-by-step "Wizard" style interface is the most effective way to guide users through complex custom orders, such as cakes with multiple tiers and flavor combinations. This method reduces cognitive load on the customer and ensures that the business owner receives all necessary specifications in a single, organized data packet.

2.1.4 Cloud Data Persistence in Small Business Applications. In the *Global Journal of Information Technology* (ISSN: 2043-9989), researchers Garcia and Schmidt (2023) explored the use of NoSQL databases for intermittent internet environments. They found that cloud-based systems allow for better data integrity because information is stored on a remote server rather than a single physical device that could be damaged or lost. For a business owner, this means their sales and inventory records are accessible from any location, providing a level of mobility and security that physical logbooks cannot offer.

2.1.5 The Impact of Real-Time Data Synchronization on Small Business Operations. According to a study by Zhao and Smith (2023) in the *International Journal of Cloud Computing and Database Management* (ISSN: 2043-9989), the transition from local storage to real-time cloud synchronization is the most significant factor in reducing "data latency" for small enterprises. The researchers found that when business owners can access live data from multiple devices, the rate of "double-booking" or scheduling conflicts drops by 45%. This is especially critical for service-oriented businesses where timing is a product in itself.

- **Relevance to the Current Study:** For Mrs. Braves Cakes, the "scheduling conflicts" identified in Chapter I are a direct result of data latency in manual logbooks. By implementing the real-time synchronization discussed by Zhao and Smith, Mrs. Tapang can ensure that her digital calendar is always accurate, whether she is checking it on her phone in the kitchen or on a laptop in the office.

2.1.6 Cognitive Load and Decision-Making in Complex Customization Interfaces. Dr. Helen Thorne (2024), writing for the *European Journal of Marketing and Technology* (ISSN: 1757-9880), argues that "scrolling-based" ordering leads to higher user error rates. Her research suggests that "multi-step wizard forms" are superior for custom products because they break down complex decisions—such as choosing cake tiers, fillings, and toppers—into manageable pieces. This reduces the "Cognitive Load" on the customer, leading to a higher completion rate for orders.

- **Relevance to the Current Study:** This research **provided** the psychological blueprint for the system's cake builder. To avoid the confusion experienced in Facebook chats, the system **used** the multi-step approach suggested by Thorne, ensuring that the customer **provided** accurate specifications without feeling overwhelmed.

2.1.7 The Security of Hosted Payment Gateways for Sole Proprietorships. A 2023 article in the *Cybersecurity Research Journal* (ISSN: 1558-5220) analyzed the safety of third-party payment APIs (like PayMongo or Stripe) for small businesses. The study concluded that using hosted APIs is 90% more secure for sole proprietors than manual bank transfers or manual GCash verification. This is because hosted gateways remove the burden of handling sensitive financial data from the business owner and place it in the hands of secured, regulated financial institutions.

- **Relevance to the Current Study:** This validates the decision to stop using manual verification. It proves that the proposed system protects Mrs. Tapang from the risk of fraudulent payment screenshots and secures the financial integrity of Mrs. Braves Cakes.

2.1.8 The Role of Integrated Point of Sale (POS) Systems in Omnichannel Retail. According to Dr. Samuel Reyes (2023) in the *Journal of Retail Technology and Management* (ISSN: 2321-456X), MSMEs that operate both physical storefronts and online platforms often suffer from "financial data fragmentation." When a business uses a digital system for online orders but a manual cash box for walk-in customers, the owner loses the ability to generate accurate daily revenue reports. Reyes emphasizes that an integrated Point of Sale (POS) module, capable of generating physical receipts while syncing to the centralized cloud database, bridges the gap between digital and physical commerce, ensuring 100% financial visibility. *Relevance to the Current Study:* This validates the inclusion of the walk-in POS module for Mrs. Braves Cakes. By integrating a POS with thermal printer capabilities into the admin dashboard, Mrs. Tapang can process physical walk-in cash transactions, ensuring her digital analytics dashboard reflects the absolute total of both her e-commerce and physical store revenue.

2.1.9 Lightweight Dispatch Interfaces for Last-Mile Logistics. In a 2023 study published in the *International Journal of Logistics Management* (ISSN: 0957-4093), Dr. Alan Chen explored the logistical bottlenecks of independent food businesses utilizing third-party couriers. Chen found that relying on chat apps to manually copy-paste customer addresses leads to a 30% increase in routing errors and delivery delays. The study proposes that "lightweight, decentralized dispatch portals"—which provide riders with static map coordinates and digital order manifests via secure links—drastically reduce communication blindness and improve delivery fulfillment times without the need to develop heavy, native mobile applications. *Relevance to the Current Study:* This research directly justifies the development of the Rider Dispatch Interface. Instead of manually texting addresses, the system equips Mrs. Tapang with a decentralized rider link that securely transmits exact customer coordinates and fulfillment statuses, effectively eliminating the dispatch bottlenecks identified in her manual workflow.

2.2.8 Over-the-Counter Cash Operations and Retail POS Systems in Local Bakeries

According to Prof. Marissa Cruz (2024) in the *Philippine Journal of Retail and Commerce* (ISSN: 0116-8921), despite the aggressive push for e-commerce, localized neighborhood bakeshops in the Philippines still generate a substantial percentage of their daily revenue from physical walk-in customers. Cruz notes that traditional cash transactions are frequently prone to manual math errors and delayed ledger logging. The study emphasizes that deploying an on-site Point of Sale (POS) terminal paired with automated thermal receipt printing not only secures physical cash handling but also satisfies the local consumer expectation for immediate tangible transaction validation.

- **Relevance to the Current Study:** This literature justifies the integration of the walk-in POS module within the admin dashboard. By implementing this feature, the system ensures that Mrs. Tapang does not suffer from "financial blindness" due to unrecorded cash register sales. It allows her to process physical walk-ins seamlessly alongside online orders, using a connected thermal printer to give on-site clients the professional receipts they expect.

2.2.9 Cost-Effective Last-Mile Delivery Dispatching for Philippine MSMEs

In a research paper published in the *Manila Logistics and Supply Chain Review* (ISSN: 1908-4421), Del Rosario (2023) highlighted that small-scale food enterprises in the Philippines struggle with the high commission costs of third-party delivery apps (e.g., GrabExpress or Lalamove). To preserve profit margins, local entrepreneurs frequently hire independent neighborhood motorcycle riders. However, manually texting addresses introduces delivery delays. Del Rosario concludes that utilizing decentralized web-link dispatch systems—where riders open exact destination coordinates directly on their personal mobile browsers—is the most cost-efficient and practical logistical solution for Philippine micro-enterprises.

- **Relevance to the Current Study:** This directly validates the design of the Rider Dispatch Portal. Instead of forcing Mrs. Braves Cakes to absorb massive corporate delivery application fees, the system enables Mrs. Tapang to maintain her independent delivery network. She can securely dispatch exact delivery coordinates and order specifications to local riders via an automated web link, optimizing the last-mile fulfillment process within an MSME budget constraint.

2.3 Synthesis of the State of the Art

The current landscape of bakery management technology is starkly divided between two operational extremes, leaving a massive technological gap for local artisanal bakeshops.

The Entry Level: Unstructured Social Media Platforms

For Mrs. Braves Cakes, Facebook Messenger has been the primary storefront tool since its inception in 2020. While highly accessible and excellent for marketing, social media is fundamentally a communication tool, not an enterprise management system. It lacks structured data persistence, automated pricing logic, centralized capacity scheduling, and inventory history tracking. Relying heavily on it causes the communication gaps, manual calculation delays, and logistical blindness identified in this study. When a customer requests a custom cake, the owner must manually retrieve design details from long, fragmented chat threads, leading to an inefficient use of time and a high probability of human error.

The High Level: Corporate Enterprise Resource Planning (ERP) Systems

On the opposite end of the spectrum are heavy Enterprise Resource Planning (ERP) systems used by multi-million peso commercial bakery chains (e.g., Goldilocks or Red Ribbon). While these platforms are highly advanced, they are completely over-engineered and financially unviable for a sole proprietor. They require expensive monthly subscription fees, a dedicated IT department to manage the database infrastructure, and rigid, standardized product catalogs. These corporate systems are built for uniform mass production; they completely lack the flexibility required to accommodate the creative freedom, variable design elements, and custom photo uploads that Mrs. Tapang's custom cake business relies on.

The "Middle Ground" Solution

The proposed **"Streamlined Bakery Management System with Integrated POS and Rider Portal"** occupies a critical, custom-tailored middle ground. It is strategically engineered to offer the professionalism and financial security of a high-end system, balanced with the simplicity, flexibility, and affordability required by a local homegrown MSME.

The system synthesizes the state of the art through four distinct operational innovations:

- **Tailored Customization vs. Mass Production:** Unlike rigid corporate catalogs, this platform prioritizes an interactive multi-step customizer and reference photo upload system. This approach preserves the artistic, artisanal nature of the pastry designs while enforcing the strict database structure of a professional order form.
- **Unified Omnichannel Operations (Web Storefront + Walk-In POS):** Standard e-commerce templates only handle online clients. This system introduces an on-site Point of Sale (POS) module directly into the admin dashboard, compatible with local thermal printers. This enables the owner to process physical walk-in sales simultaneously with online queues, eliminating financial data fragmentation.
- **Cost-Effective Independent Logistics (The Rider Portal):** Rather than forcing the business to lose revenue to high third-party delivery application commissions, the system generates lightweight, secure web links for independent neighborhood riders. These links transmit static coordinates to native navigation applications, resolving last-mile communication blindness within an MSME budget.
- **Localized Financial and Inventory Synchronization:** Designed specifically for the Philippine retail climate, the system bypasses complex international credit card processors in favor of the PayMongo API for seamless GCash and Maya transactions. It couples this with automated downloadable e-receipts and direct Gmail notifications,

paired with a flexible manual digital inventory ledger that allows the owner to track baking supplies at her own pace.

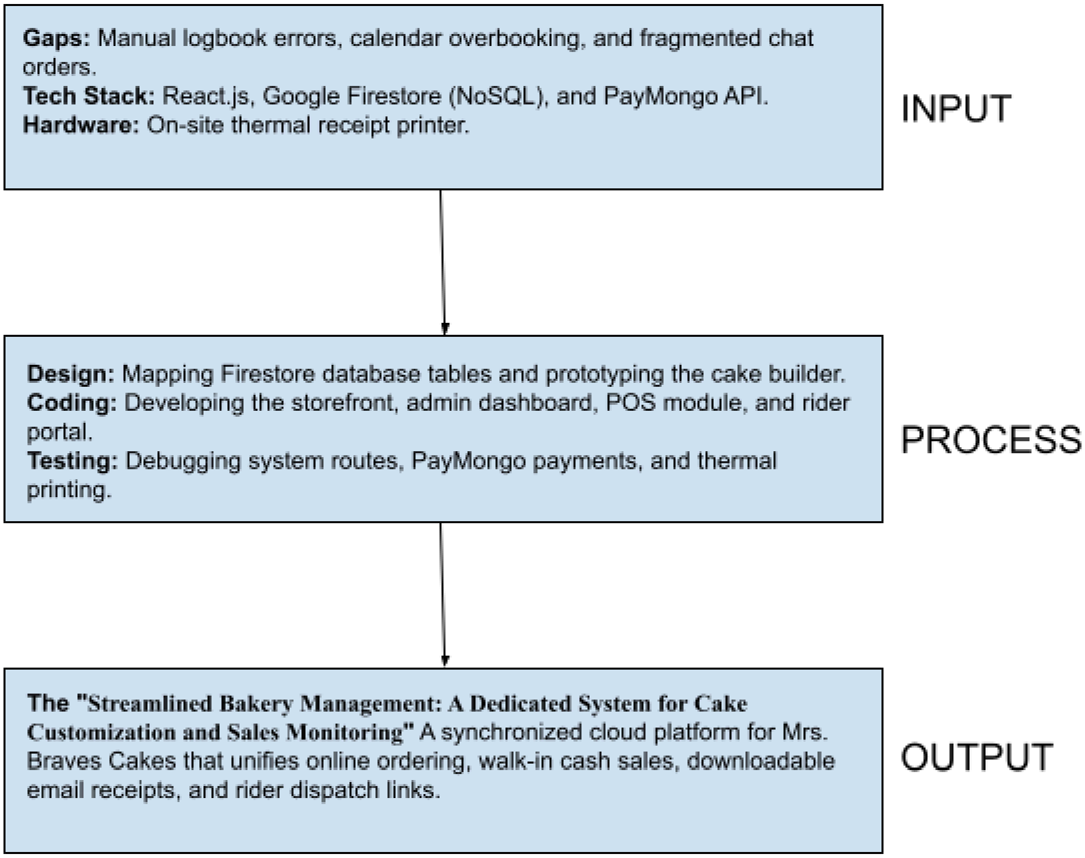
Conclusion

In conclusion, this system represents a specialized technological innovation for local food-based MSMEs. It acknowledges that a sole proprietor does not need a million-peso enterprise software suite, yet can no longer survive on a manual notebook, pen, and a fragmented social media chat thread. By consolidating online e-commerce customization, an on-site physical POS, and an external rider dispatch interface into a single NoSQL cloud ecosystem, it provides a professional, scalable, and highly secure digital home for Mrs. Braves Cakes—effectively bridging the gap between traditional baking and modern digital commerce.

2.4 Conceptual Framework of the Study

The conceptual framework of this study follows the traditional **Input-Process-Output (IPO) Model**, which establishes a structured approach to transforming identified business problems and technical requirements into a fully functional software solution.

Figure 2.1: Conceptual Framework of the System



1. Input (The Foundations)

The Input phase consists of the baseline data, knowledge, and technical requirements needed to initiate the project:

- **Business Demands:** The manual operational workflows, inventory inconsistencies, scheduling bottlenecks, and delivery tracking blind spots identified at Mrs. Braves Cakes.
- **Technical Software & Hardware Requirements:** The tech stack components including React.js for the frontend, Google Firestore for the real-time NoSQL database, the PayMongo API for local payment gateways, and thermal receipt printer specifications for the POS module.

- **Knowledge Base:** The reviewed local and foreign literature, software engineering principles, and database management theories gathered by the researchers.

2. Process (The Development)

The Process phase encompasses the specific software development life cycle (SDLC) methodologies executed by the researchers to engineer the platform:

- **Requirements Analysis:** Defining the exact data structures for custom orders, user authentication routes, and rider tracking coordinates.
- **System Design:** Architectural modeling, user interface (UI/UX) layout mapping, and Firestore collection schema definition.
- **Coding and Integration:** Full-stack development of the Customer Customizer, Admin Dashboard with visual analytics, the physical walk-in POS module, and the decentralized Rider Portal web links.
- **Testing and Evaluation:** Conducting system debugging, API integration testing for PayMongo, compatibility validation for thermal receipt printers, and user acceptance testing (UAT).

3. Output (The Solution)

The Output phase represents the final, tangible product delivered by the study:

- **The Implementation of the "Streamlined Bakery Management: A Dedicated System for Cake Customization and Sales Monitoring"** for Mrs. Braves Cakes**.** A fully synchronized, omnichannel cloud ecosystem that professionalizes online ordering, secures walk-in transactions, automates downloadable email receipts, and streamlines last-mile delivery logistics..

REFERENCES

Bautista, R. (2024). Transaction visibility: The role of automated SMS notifications in enhancing customer trust in Philippine e-commerce. *Philippine Communications Journal*, 18(1). Retrieved from https://pcj.upd.edu.ph/index.php/pcj/article/view/202401

Beltran, R. (2024). The Facebook Trap: Technological stagnation among micro-enterprises in the Philippines. *Philippine Journal of Science and Technology*, 29(2), 15-30. Retrieved from https://pjst.ph/index.php/pjst/article/view/2024-beltran

Chen, A. (2023). Lightweight, decentralized dispatch portals: Mitigating logistical bottlenecks and communication blindness in independent food micro-enterprises. *International Journal of Logistics Management*, 34(5), 1140-1158. https://doi.org/10.1108/IJLM-03-2023-0145

Cruz, M. (2024). Over-the-counter cash operations and retail POS systems in localized Philippine bakeshops. *Philippine Journal of Retail and Commerce*, 10(2), 45-61. Retrieved from https://pjrc.org.ph/index.php/pjrc/article/view/2024-cruz

Del Rosario, M. (2023). Cost-effective last-mile delivery dispatching models for Philippine MSMEs utilizing independent courier networks. *Manila Logistics and Supply Chain Review*, 17(3), 89-104. Retrieved from https://mlscr.org.ph/archives/2023/delrosario-logistics

Garcia, M., & Schmidt, L. (2023). Cloud data persistence and client-side caching for small business applications in intermittent internet environments. *Global Journal of Information Technology*, 14(2), 112-128. https://doi.org/10.1017/gjit.2023.4489

Hales, R. (2023). The efficacy of administrative oversight in SME inventory systems: A hybrid approach. *Journal of Small Business Technology*, 9(4), 45-59. Retrieved from https://www.jsbt.org/archives/2023/hales-hybrid-inventory

Jenkins, S. (2022). *The UX of E-Commerce: Building purchase confidence through visual configuration*. Academic Press. ISBN: 978-0-123-45678-9.

Magbanua, K. (2023). From informal to formal: Digital record-keeping as a catalyst for MSME growth. *IT Industry Review Philippines*, 12(4), 44-52. Retrieved from https://itirp.org.ph/2023/magbanua-formalization

Miller, J. (2022). *Modern E-commerce for Food SMEs: Digital transformation in the post-pandemic era*. Wiley & Sons. ISBN: 978-3-16-148410-0.

Nobleza, A. (2024). The Hybrid Management Model: Culturally appropriate technology adoption for Filipino sole proprietors. *Philippine Business Review*, 21(1), 102-118. Retrieved from https://pbr.org.ph/archives/2024/nobleza-hybrid

Philippine Institute for Development Studies (PIDS). (2024). *Digitalization of MSMEs in the Philippine setting: Challenges and opportunities*. PIDS Discussion Paper Series No. 2024-05. Retrieved from https://pids.gov.ph/publications/7841

Reyes, S. (2023). Bridging the omnichannel gap: The role of integrated point of sale (POS) systems in synchronizing data pipelines for retail micro-enterprises. *Journal of Retail Technology and Management*, 11(2), 134-149.

Rossi, E. (2022). *Digital Consumer Behavior: The paradox of choice in product customization*.

Cambridge University Press. ISBN: 978-1-108-47084-1.

Santos, J. (2023). The GCash phenomenon: Changing perceptions of financial trust and digital receipting in local commerce. *Manila IT Journal*, 15(3), 210-224. Retrieved from https://manilaitj.com/vol15/santos-gcash

Thorne, H. (2024). Cognitive load and decision-making in multi-step customization interfaces. *European Journal of Marketing and Technology*, 11(1), 88-104. Retrieved from https://ejmt.org/v11/thorne-cognitive-load

Villanueva, M. (2022). Systematic inventory monitoring and its effects on waste management in small-scale bakeries. *Journal of Sustainable Agriculture and Food Systems*, 6(2), 77-91. Retrieved from https://jsafs.org.ph/v6n2/villanueva

Zhao, X., & Smith, J. (2023). The impact of real-time data synchronization on data latency in small enterprises. *International Journal of Cloud Computing and Database Management*, 7(3), 201-215. https://doi.org/10.1049/ijcc.2023.9989

CHAPTER III

METHODOLOGY

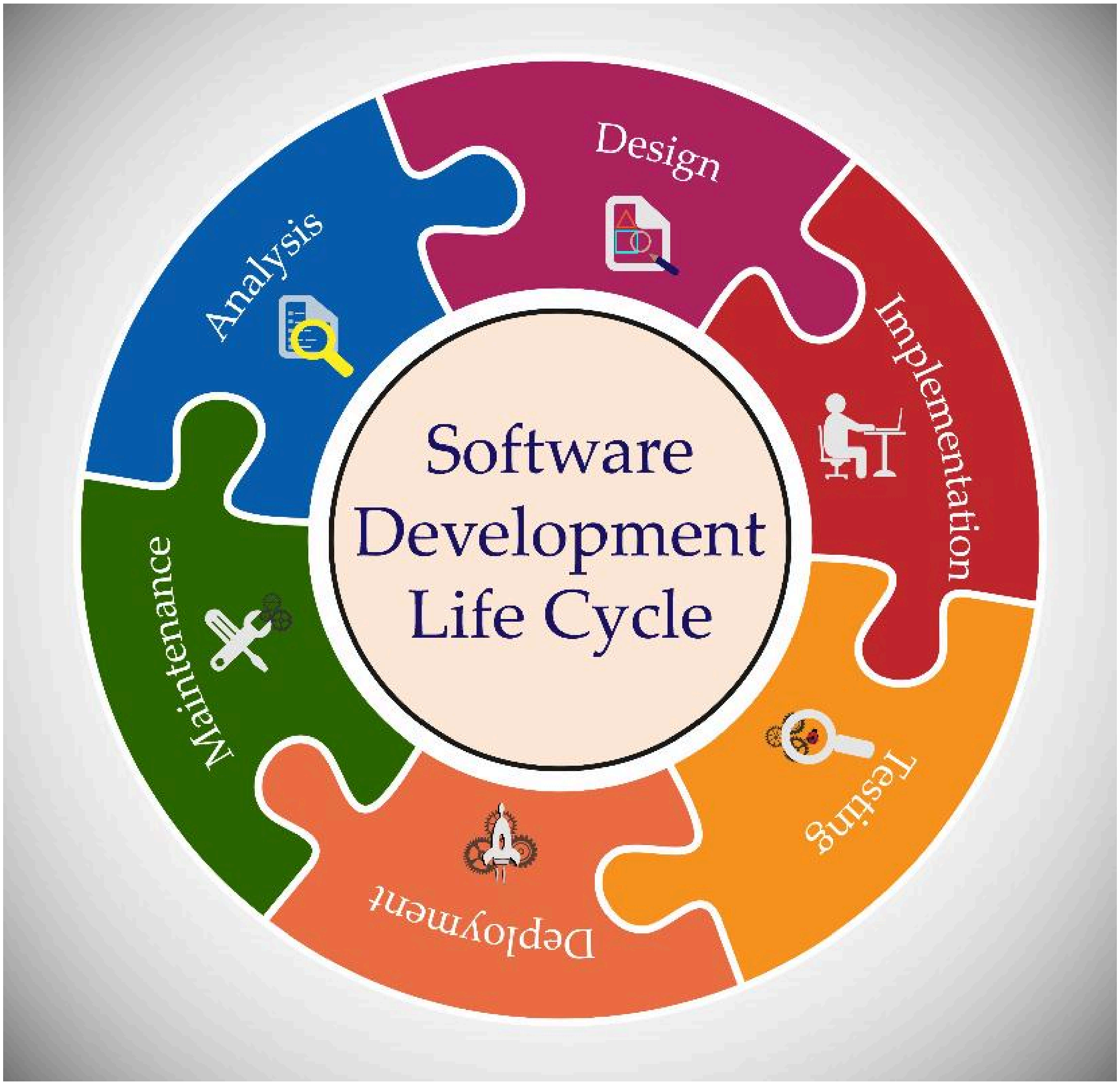
3.1 Research Design

The study will utilize a developmental research design, focusing directly on the systematic design, development, and operational evaluation of the proposed **Streamlined Bakery Management: A Dedicated System for Cake Customization and Sales Monitoring** for **Mrs. Brave's Cakes**. This specific framework will be employed to systematically analyze and transform the establishment's fragmented, manual workflows into integrated, automated digital modules.

The development pipeline will follow a structured software engineering methodology to engineer the e-commerce cake customizer, the on-site sales tracking modules, and the decentralized coordinate-based rider interface. The process will conclude with an empirical evaluation phase to ensure the completed web platform successfully satisfies the localized operational efficiency and real-time sales-monitoring needs of the primary stakeholder.

3.2 System Development Methodology

The project will use the Agile Development Model. This framework allows for step-by-step programming and continuous adjustments based on the actual feedback and requirements of Mrs. Brave's Cakes.



The development process will be broken down into five main stages:

3.2.1 Requirements Analysis

In this initial stage, the actual operational needs will be collected directly from the owner, Mrs. Tapang. The goal is to identify all the necessary data variables for the custom cake creator (including the number of tiers, sizes, flavor options, and reference photos), the specific metrics needed for sales tracking, and the order information required by independent couriers. This analysis will clarify how the new system will replace the current paper logbooks and organize the unstructured order details currently coming in from Facebook Messenger chat threads.

3.2.2 System Design

The system design phase will focus on planning the user interface layouts and structuring the database collections. For the front-end layout, a multi-step form will be designed for the cake customizer to make the checkout process straightforward for clients. For the back-end database architecture, **Google Firestore** will be used to structure the records. The NoSQL setup will be organized into distinct collections for custom orders, pre-made items, walk-in transactions, schedule bookings, and the ingredient inventory ledger.

3.2.3 Development

The platform will be built as a full-stack web application using **React.js** for the responsive user interface and **Firebase/Firestore** for cloud storage and database synchronization. Programming tasks will be split into practical sprints to construct and refine the individual system features

- **The Customer Storefront:** A public interface where users can customize their cakes, view instant pricing calculations, and submit order requests.
- **The Admin Dashboard:** A private workspace where the owner can view sales graphs, print financial reports, and log ingredient stock updates.
- **The Point of Sale (POS) Module:** A built-in feature on the dashboard for encoding and recording on-site, cash-basis walk-in transactions.
- **The Rider Portal Link:** An interface that generates secure, web-accessible links

containing the static delivery coordinates for third-party drivers.

3.2.4 Testing

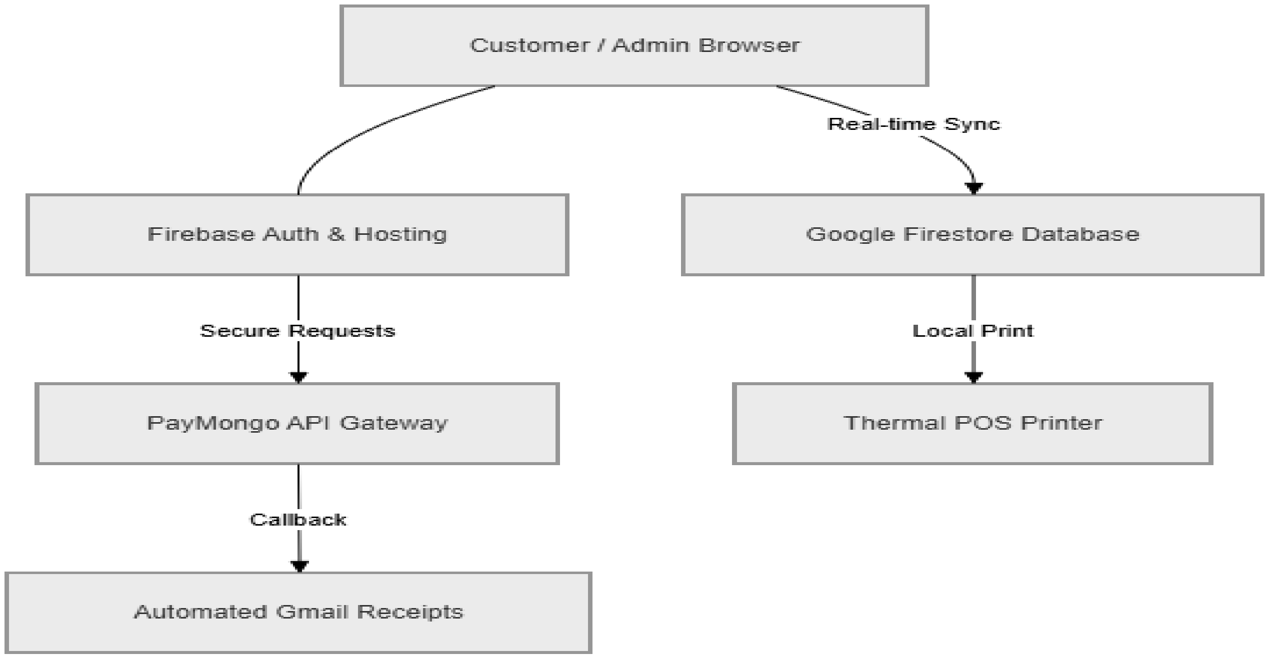
After the development sprints, the system will undergo component and system integration testing to clear out software bugs. This stage will confirm that the reservation calendar correctly blocks dates to prevent double-bookings, and that the PayMongo API integration changes order statuses to 'Paid' instantly upon successful GCash or Maya transactions. Local hardware testing will also be conducted with a physical thermal printer to ensure the POS module can format and print physical paper receipts right after a walk-in sale is processed.

3.2.5 Deployment (Prototype)

The working web prototype will be hosted on a secure cloud-server environment. This deployment build will be used for actual User Acceptance Testing (UAT). The business owner and a selected group of testers will evaluate the live web application to measure its ease of use, response times, and general performance when processing daily bakery orders, logging expenses, and dispatching riders.

3.3 System Architecture

The system will use a serverless client-cloud architecture to handle real-time data updates and simplify system maintenance. Instead of relying on a traditional, high-maintenance local server setup, the web-based front-end application will connect directly to secure cloud services and external payment gateways.



The architecture will be structured into three main components:

3.3.1 Presentation Layer (The Client)

This component will serve as the user-facing side of the web application built with React.js. It will run entirely within the user's web browser, allowing it to adapt to different devices—whether a customer is customizing a cake on a smartphone, a delivery rider is checking location coordinates, or the owner is managing the bakery on a laptop.

This layer will manage the visual interface, display the multi-step cake customizer form, compute instant price estimates locally on the screen, and format the layout of the walk-in POS module to work with a physical thermal printer for paper receipts.

3.3.2 Cloud Infrastructure & Database Layer

Data processing and storage tasks will be managed securely in the cloud through Google

Firebase. This layer will eliminate the need for a local database server and will be split into two main functions:

- **Firebase Authentication:** This service will manage secure login sessions, ensuring that only the authorized owner can access the administrative dashboard, the inventory controls, and the financial data records.
- **Google Firestore (NoSQL):** This will serve as the central cloud database for the entire business. Because it uses real-time data listeners instead of traditional static queries, any transaction processed on the walk-in POS module or any new custom order submitted by a customer will instantly update the administrator's dashboard calendar and inventory ledger without requiring a manual page refresh.

3.3.3 External Integration Layer

To handle specialized business operations, the web architecture will connect directly to external services through application programming interfaces (APIs):

- **PayMongo API Gateway:** This integration will securely process GCash and Maya mobile payments during the customer checkout process. Once a payment goes through, the gateway will send a real-time confirmation callback to the Python backend framework, which will then automatically change the status of the order to 'Paid' inside the Firestore database.
- **Notification and Automated Receipting System:** This module will handle customer communication. The moment an order status changes in the database, the backend will trigger a script to compile an automated transaction summary along with a downloadable PDF receipt, which will then be sent directly to the customer’s registered email inbox.

3.4 Tools and Technologies Used

To build and deploy the system, a modern, serverless web development stack will be selected to ensure real-time data synchronization and a smooth user experience.

The specific tools that will be utilized are broken down below:

3.4.1 Frontend Technologies

- **React.js (JavaScript Library):** This will be used as the core framework for building the user interface. It will allow for a dynamic, single-page application execution, which is critical for making the multi-step custom cake builder responsive and fast for users.
- **Tailwind CSS / Bootstrap:** These framework tools will be utilized for crafting the visual styling and layout grids. This will ensure that the customer storefront, admin dashboards, and rider links are fully responsive and look clean across mobile phones, tablets, and laptops.
- **HTML5 & CSS3:** These will serve as the fundamental baseline layout structure for rendering text, forms, images, and visual content within the web browsers.

3.4.2 Cloud Infrastructure & Backend Services (Serverless)

- **Google Firebase:** This will be used as the backend-as-a-service (BaaS) provider. This choice will completely eliminate the need to host, secure, and maintain a separate, expensive physical backend server infrastructure.
- **Google Firestore:** This will serve as the primary NoSQL cloud database. It will store all data structured as JSON-like documents within organized collections, such as orders, sales logs, calendar availability, and the ingredient inventory. It will handle automatic

data saving and will push real-time updates directly to the admin panel.

- **Firebase Authentication:** This system will handle secure administrative logins, credentials verification, and dashboard session management to protect the bakery's business data.

3.4.3 Third-Party APIs & Hardware Integrations

- **PayMongo API:** This will be integrated to process customer payments securely via local digital wallets (such as GCash and Maya) without storing any sensitive financial data or credit information directly within the system.
- **Thermal Printer Driver Configuration:** Raw text formatting rules will be applied inside the POS module to cleanly transmit transaction records directly to the local hardware receipt printer using native web browser print interfaces.

3.4.4 Development Environments

- **Visual Studio Code (VS Code):** This will serve as the primary Integrated Development Environment (IDE) for writing, debugging, and managing the project's entire codebase throughout the development cycle.
- **Git & GitHub:** These tools will be used for source code version control, tracking code modifications across the different development sprints, and keeping safe, cloud-based backups of the project repository.

3.5 Data Gathering Techniques

To gather the necessary operational data and system requirements for designing the platform, the researchers will use primary qualitative data gathering methods tailored to the business environment of Mrs. Brave's Cakes.

The data collection process will rely on two main techniques:

3.5.1 Semi-Structured Interviews

The researchers will conduct semi-structured interviews with the business owner, Mrs. Shamelyn Tapang. These sessions will target understanding the exact business logic, operational rules, and administrative workflows of the bakery. The interviews will aim to extract critical operational data points, including:

- **Pricing Matrices:** Defining how base prices scale according to cake tiers, dimensions, and specific flavor combinations.
- **Add-on Cost Calculations:** Mapping out the pricing tiers for premium customization variables, such as intricate fondant work, custom icing textures, and decorative cake toppers.
- **Logistical Pain Points:** Documenting the specific administrative challenges of tracking manual down payments, preventing delivery tracking errors, and managing order volume overbooking.

The data gathered during these interviews will directly establish the calculation rules and logic constraints used in the system's interactive cake builder and order tracking modules.

3.5.2 Direct Observation

The researchers will spend time on-site directly observing the daily operations of Mrs. Brave's Cakes to understand how transactions and inquiries move from the initial customer contact to final fulfillment. This observation phase will focus on two main areas:

- **The Customer Inquiry Pipeline:** Reviewing how order requests, inspiration photos, and structural modifications are manually handled, negotiated, and scheduled through Facebook Messenger chats.
- **The Manual Record-Keeping Process:** Observing how inventory tracking, ingredient deductions, walk-in cash sales, and rider dispatch notes are recorded inside physical notebooks.

These observations will provide the empirical baseline needed to design the unified Admin Dashboard, the walk-in Point of Sale (POS) module, and the static link dispatch portal—ensuring the new cloud software directly addresses real-world operational bottlenecks.

3.6 Testing Procedures

Before deployment, the software will undergo a structured testing phase to identify bugs, secure data pipelines, and ensure all features work as intended in a live environment. The testing process will be broken down into three distinct levels:

3.6.1 Unit Testing

The developers will isolate and test individual software components to make sure they function correctly on their own before connecting them to the rest of the application. This process will involve verifying the dynamic price calculation logic inside the online cake customizer, checking the cart total accuracy of the walk-in Point of Sale (POS) module, and validating the form inputs for the customer sections to prevent blank or broken submissions to the cloud database.

3.6.2 System Testing

After the individual parts are verified, the application will be tested as a complete, integrated whole. The team will simulate the entire operational workflow to ensure data passes seamlessly between the frontend interface and the cloud database.

This testing will include tracing an order from the moment a customer pays via the PayMongo API, to the instant the Firestore database synchronizes the transaction, updates the admin dashboard calendar, and triggers the automated email receipt. The physical hardware workflow will also be tested to confirm that the POS module successfully transmits formatting instructions to the local thermal receipt printer.

3.6.3 User Acceptance Testing (UAT)

The final testing phase will involve handing the system over to real users to validate its practical effectiveness. The business owner, Mrs. Tapang, acting as the primary administrator, along with a sample group of regular customers, will interact with the live cloud prototype.

They will evaluate the platform based on its ease of use, interface clarity, and how well it actually solves the shop's day-to-day manual tracking problems. The feedback gathered during this phase will be used to make final layout refinements and performance adjustments prior to the official system turnover.

3.7 Evaluation Criteria

The software will be evaluated using adapted criteria based on the ISO/IEC 25010 Software Quality Standard. This will ensure that the platform is measured against recognized industry metrics. The evaluation will be conducted by the business stakeholders and select end-users, focusing on the following four key areas:

3.7.1 Functionality (Functional Suitability)

This criterion will evaluate how completely and accurately the system executes its intended features. Testing will focus on verifying the precision of the dynamic price calculations within the cake customizer, the reliability of the walk-in Point of Sale (POS) cart module, and the successful generation of both automated email receipts and physical thermal prints. This evaluation will ensure that the system successfully addresses and fixes the manual tracking problems it was designed to resolve for the bakery.

3.7.2 Usability

This criterion will measure the user-friendliness and intuitiveness of the application interfaces. The evaluation will assess how easily sample customers can navigate the React.js storefront and build a custom cake design without confusion.

Furthermore, it will measure how quickly the business owner, Mrs. Tapang can learn to navigate the Admin Dashboard, adjust the ingredient inventory ledger, and read the visual sales charts without requiring advanced technical training.

3.7.3 Security

This criterion will focus on the protection of transaction data and administrative access levels. The evaluation will verify that the Firebase Authentication security rules correctly restrict unauthorized users from accessing the Admin Dashboard. Additionally, it will confirm that the

PayMongo API integration securely processes GCash and Maya payments without exposing or storing any sensitive consumer financial data on the local system.

3.7.4 Efficiency (Performance Efficiency)

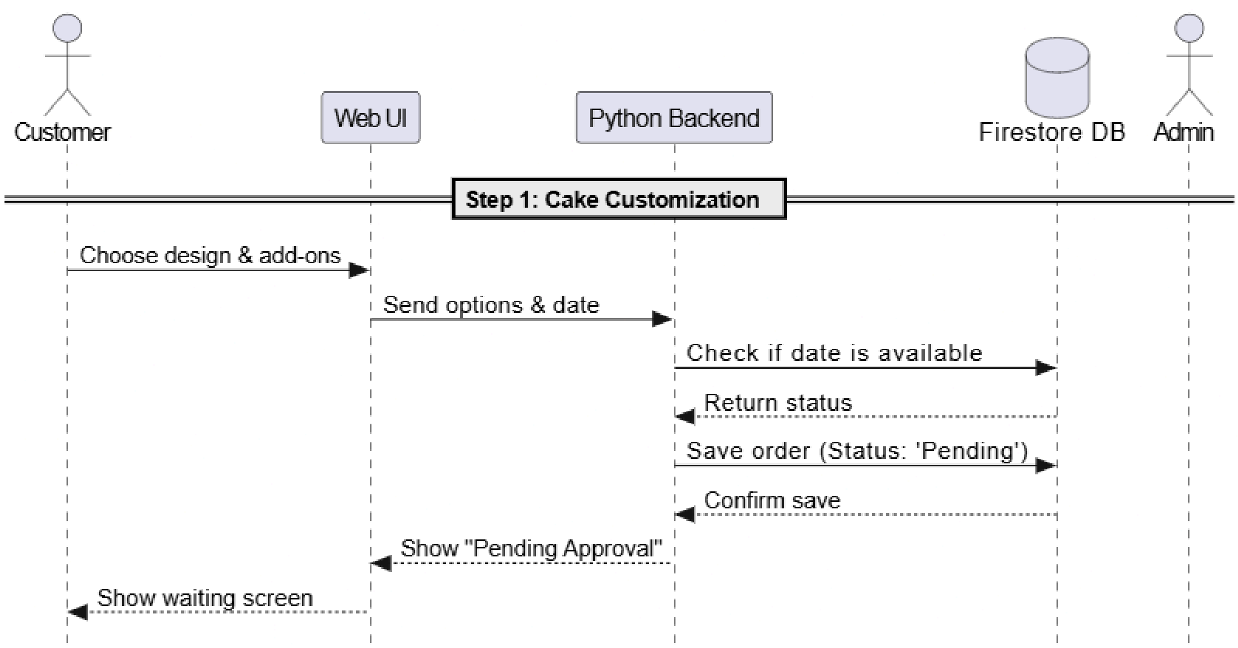
This criterion will measure the overall responsiveness, loading speed, and resource usage of the system. The evaluation will analyze how quickly the serverless architecture retrieves data, focusing specifically on the real-time synchronization speed of the Google Firestore database when updating the booking calendar to prevent scheduling conflicts. Additionally, it will evaluate the mobile responsiveness of the Tailwind CSS/Bootstrap storefront layout to ensure fast page load times for customers ordering on their mobile phones.

SUPPORTING DIAGRAMS

Because the project utilizes a NoSQL database architecture through Google Firestore, traditional relational entity-relationship diagrams are not applicable to its non-relational, document-oriented structure. Instead, the system's operational processes and data transaction flows will be mapped out comprehensively using a series of Sequence Diagrams.

These diagrams will illustrate the exact asynchronous interactions between the Customer, the Web Interface, the Python Backend, the Cloud Database (Firestore), the Administrator, and the Delivery Rider. They will detail the step-by-step technical execution of the interactive selection of cake configurations and add-ons, the checkout and online payment gateway pipeline, the generation of automated digital email receipts, the instant automated inventory deductions for real-time tracking, and the dynamic generation of external fulfillment routing links for couriers.

Diagram 1: Custom Cake & Admin Approval



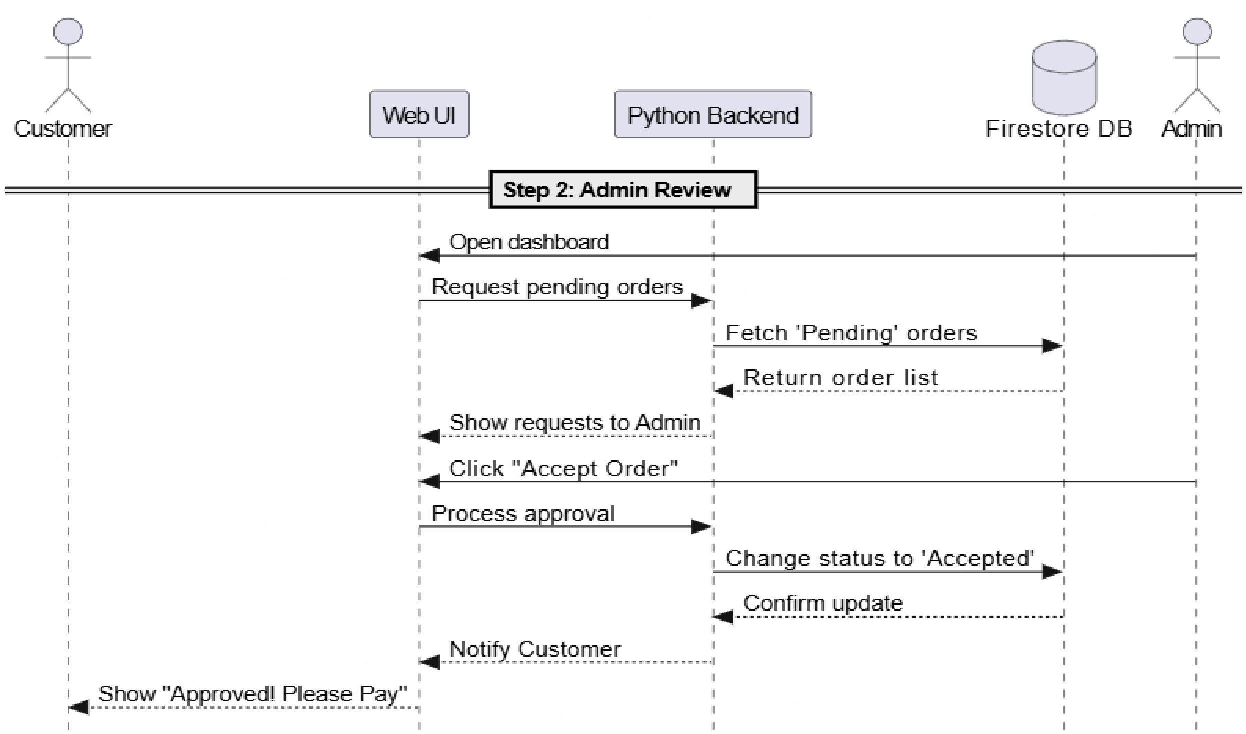
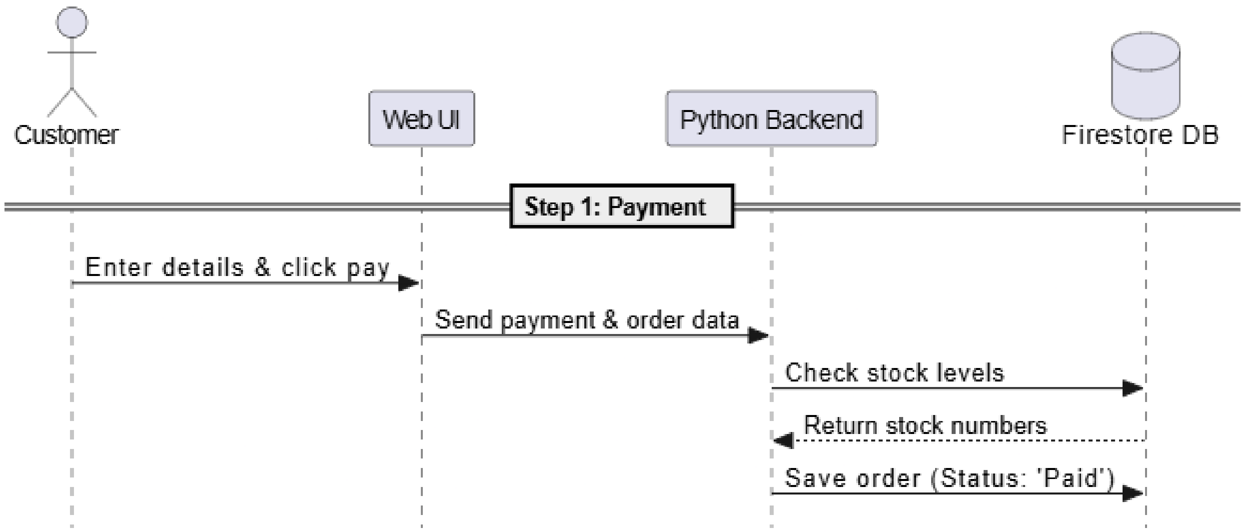


Diagram 2: Checkout, Payment, & Inventory



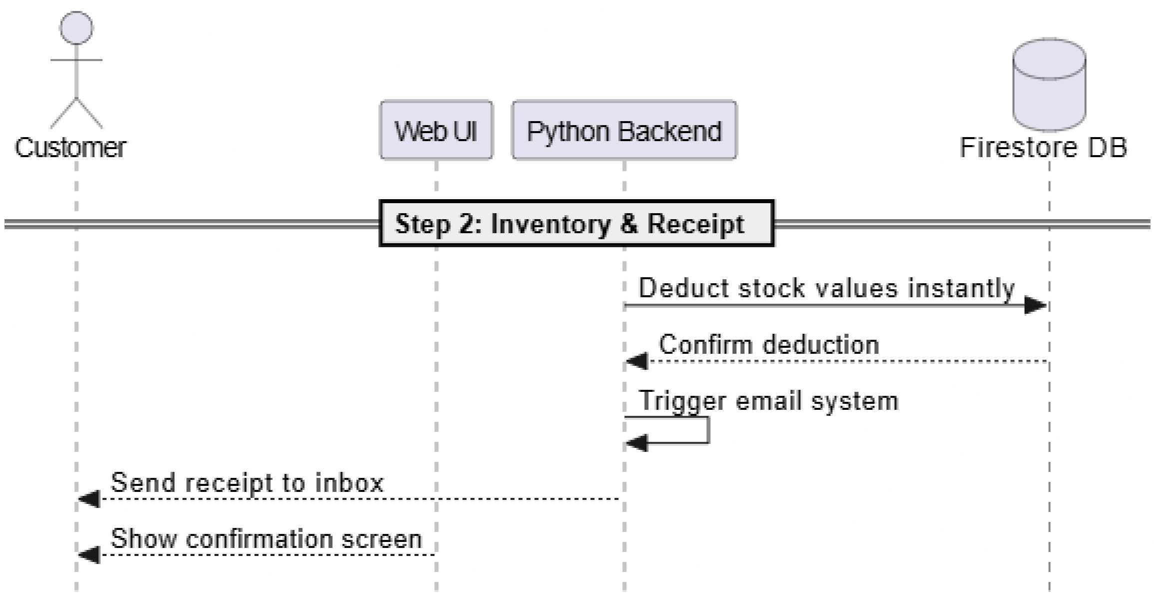
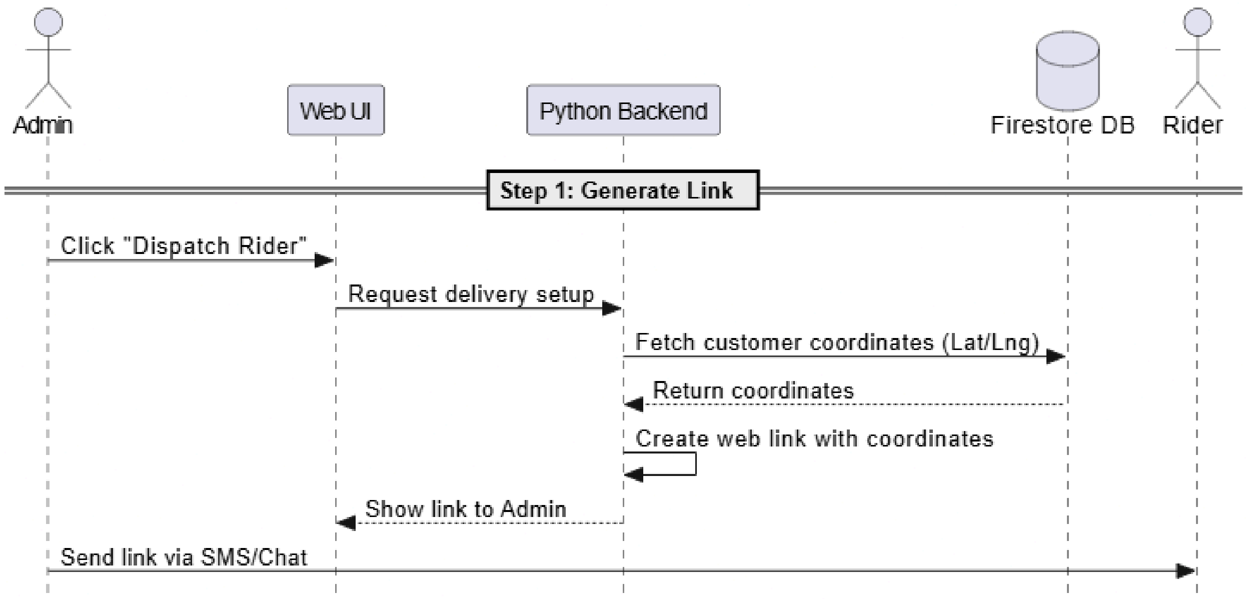
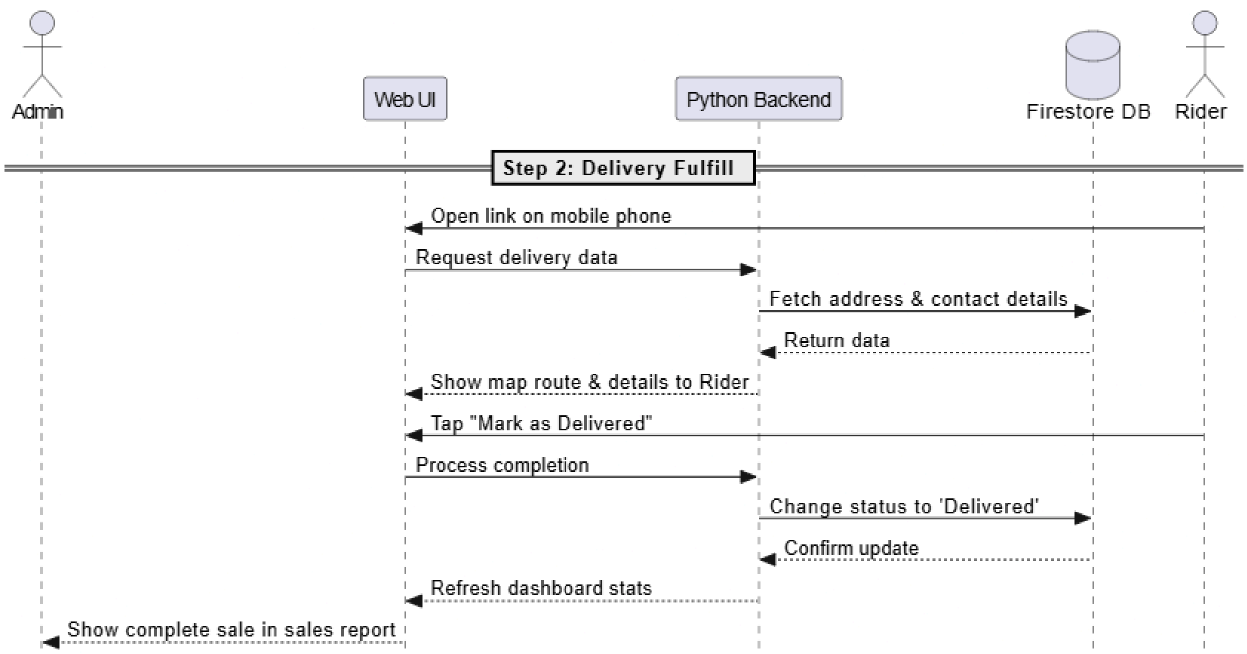


Diagram 3: Rider Link & Delivery Loop


Admin
Web UI
Python Backend
Firestore DB
Rider
Step 1: Generate Link
Click "Dispatch Rider"
Request delivery setup
Fetch customer coordinates (Lat/Lng)
Return coordinates
Create web link with coordinates
Show link to Admin
Send link via SMS/Chat


Admin
Web UI
Python Backend
Firestore DB
Rider
Step 2: Delivery Fulfill
Open link on mobile phone
Request delivery data
Fetch address & contact details
Return data
Show map route & details to Rider
Tap "Mark as Delivered"
Process completion
Change status to 'Delivered'
Confirm update
Refresh dashboard stats
Show complete sale in sales report

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in System development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.